



Bachelor study programme
Medical and Bioinformatics
A-4232 Hagenberg, Austria

A population-based multi-agent approach for simulating large-scale system dynamics in discrete environments

Bachelor thesis

to obtain the academic degree of
Bachelor of Science in Engineering

Submitted by

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Hagenberg, August 2026

Declaration

I hereby declare and confirm that this thesis is entirely the result of my own original work. Where other sources of information have been used, they have been indicated as such and properly acknowledged. I further declare that this or similar work has not been submitted for credit elsewhere.

Date

Signature

Acknowledgment

Special thanks to Kurzgesagt for inspiring the thesis' topic in their video "Why Earth Sucks Compared to the Planet Hestia".

Abstract

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Kurzfassung

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Chapter 1

Introduction

1.1 Motivation

When we look at nature, we see an incredibly complex interplay of countless animal and plant species. They prey on one another, compete for space, adapt to temperatures, and change the world around them simply by existing. This process, known as evolution, has shaped our Earth over billions of years. However, modern science faces a fundamental problem: we cannot put an entire planet into a laboratory. If we want to understand how a lifeless rock turned into a green, oxygen-rich world full of life, we run into an issue of time. Real evolution is far too slow to be observed in real-time.

This is exactly where modern bioinformatics comes into play. The overarching goal of this research field is to move evolution into the computer. If we can program realistic virtual worlds, we can simulate millions of years of Earth's history in a single afternoon. This is not just a theoretical game; it has enormous practical value for the real world. By understanding the fundamental rules of how entire ecosystems build up or collapse, we can better predict how our modern environment will react to dramatic changes—whether caused by global climate change, invasive species, or pollution. We are essentially creating a “digital twin” of ecological networks to test the breaking points of life.

This thesis focuses on developing exactly this kind of planetary simulation. The goal is to build a computational framework that can simulate the development of species on a global scale. However, instead of trying to calculate every single animal or every leaf on a tree—which would immediately overload even the most powerful supercomputers—this work introduces a clever, alternative approach. The virtual world is divided into a massive spatial grid. The cells in this grid do not contain individual animals, but rather entire populations of species acting as agents. If a species thrives, its “influence”

in that region grows. If it gets too crowded or the climate becomes too harsh, a part of the population migrates to neighboring cells.

A central element of this simulation is the realization that life does not merely adapt to its environment; it actively rebuilds it. A famous example from Earth's history, which is modeled in this work, is the Great Oxidation Event. Early plant-like life forms produced massive amounts of oxygen. What is vital for us today was pure poison for the anaerobic world back then. By simply spreading, these plants changed the chemical composition of the entire atmosphere. This creates enormous evolutionary pressure: other species must adapt, flee, or go extinct. Such massive, global feedback loops between life and the environment will be realistically modeled in the system developed here.

Furthermore, this thesis addresses a highly fascinating core question: What actually happens when life has it “too easy”? Our hypothesis is that under perfect global living conditions—meaning it is warm everywhere, there is plenty of food, and there are no extreme environmental barriers—the absolute biodiversity on the planet paradoxically decreases rather than increases. If nature does not force species to specialize in certain harsh niches, so-called “generalists” emerge. These are super-species that can survive almost anywhere. They spread unhindered across the globe and ultimately outcompete all other, weaker species. While there might be many animals in one specific location, looking at the planet as a whole, every region looks exactly the same. This theory of biological homogenization will be put to the test through the simulation.

To realize this ambitious project, standard programming is not enough. The simulation is written in the modern, high-performance programming language Rust. Additionally, to ensure that the computer knows how the millions of populations should behave realistically, advanced mathematics and techniques from the field of artificial intelligence are utilized in the background. This *AI* does not actively control the simulation during runtime; rather, it is used beforehand to translate the extremely complex biological rules into simple, highly efficient mathematical formulas.

In summary, this thesis combines biology, environmental physics, and cutting-edge software engineering. It provides a computational tool to make the complex, often invisible gears of macroevolution tangible, helping us understand why our planet's biodiversity looks the way it does today—and how quickly it can collapse when the rules of nature change.

1.2 Informatics Relevancy

1.3 Biological Relevancy

1.4 Hypothesis

α -Diversity $\sim$ γ -Diversity

Chapter 2

Foundational works

2.1 Tierra

[https://en.wikipedia.org/wiki/Tierra_\(computer_simulation\)](https://en.wikipedia.org/wiki/Tierra_(computer_simulation))

2.2 Avida

Inspired by Tierra (Chapter 2.1)

<https://de.wikipedia.org/wiki/Avida>

2.3 Räumlich erweitertes *TaNa* Modell

2.4 Multi-Level Makroevolutions-Frameworks

State of the Art, Bottom-Up Ansätze. 2025 von Latorre et al. in PLoS ONE. (Latorre et al., 2025)

2.5 klassische *ABMs*

Oft nicht skalierbar

CoMSES Net (Open*ABM*)

2.6 Homeostatic Reinforcement Learning

https://www.researchgate.net/publication/354597646_Continuous_Homeostatic_Reinforcement_Learning_for_Self-Regulated_Autonomous_Agents (Laurençon et al., 2021)

2.7 Lotka-Volterra-Dynamiken

2.8 Das “EcoTwin”-Projekt

Twin models of natural ecosystems hold great promise for informing real-world decisions on sustainable land use and biodiversity conservation. However, existing simulations of animal behavior often rely on manually crafted rules, limiting their scalability and practical utility. Here, we present a flexible and scalable agent-based modeling approach that uses reinforcement learning—instead of hand-coded rules—to simulate animal behavior. We validate this approach across ten alpine ecosystems featuring wolves, chamois, and vegetation. By comparing model outputs with empirical data, we show that the simulations reproduce realistic ecological and behavioral patterns, including population dynamics, life history traits, and social interactions. We then use the model to assess ecosystem resilience under scenarios of habitat degradation, game hunting, and heat stress. Our framework paves the way for realistic simulations advancing our ability to predict ecosystem responses to disturbance and tipping points leading to biodiversity loss, in order to support conservation planning and guide the sustainable use of natural resources.

(Strannegård et al., 2025)

2.9 “PredPreyGrass” via MADRL (CoMSES Net Repository)

Exploring learned cooperation, coevolution and free-riding. Learning is achieved through Multi-Agent Deep Reinforcement Learning (MADRL) in an ecological environment. The environment emits no other than sparse reproduction rewards. No reward shaping, no explicit cooperation signal. (HBP1969, 2026)

2.10 Multi-Agent Inverse Reinforcement Learning

Recent comparative studies between classical population-based modeling (PBM)—which typically uses differential equations—and agent-based modeling (*ABM*) have increasingly applied Multi-Agent Inverse Reinforcement Learning (MIRL). Instead of prescribing a reward function to the agents, MIRL uses empirical macroscopic data to infer, in reverse, the underlying reward functions that drive real-world collective or dispersal behaviors.

These studies provide mathematical evidence that *ABMs*, when applied to very large and homogeneous populations, can accurately approximate the same macroscopic dynamics as traditional PBM approaches. This finding directly supports the design choice of this thesis: modeling agents as aggregated population units, rather than simulating millions of individual entities, is both justified and efficient for capturing large-scale ecological dynamics.

2.11 Agent-Based Modeling vs. Population-Based Modeling

This paper addresses comparative evaluation of population-based simulation in comparison to agent-based simulation for different numbers of agents. Population-based simulation, such as for example in the classical approaches to predator-prey modelling and modelling of epidemics, has computational advantages over agent-based modelling with large numbers of agents. Therefore the latter approaches can be considered useful only when the results are expected to deviate from the results of population-based simulation, and are considered more realistic. However, there is sometimes also a silent assumption that for larger numbers of agents, agent-based simulations approximate population-based simulations, which would indicate that agent-based simulation just can be replaced by population-based simulation. The paper evaluates such assumptions by two detailed comparative case studies: one in epidemics, and one in economical context. The former case study addresses the spread of an infectious disease over a population. The latter case study addresses the interplay between individual greed as a psychological concept and global economical concepts. It is shown that

under certain conditions agent-based and population-based simulations may show similar results, but not always. 1 Parts of this paper have been presented at the conferences ECMS'08 [20] and IEA/A/E'10 [7]. The current paper extends these conference papers by providing additional simulation experiments, simulation results relating to empirical data, more detailed analyses of simulation results, and a more extensive discussion of related work and of the differences between agent-based and population-based modelling.

(Bosse et al., 2012)

Grundlage für den Ansatz, dass Populationbasierte Modelle für sehr große und homogene Populationen exakt das gleiche makroskopische Verhalten wie traditionelle PBMs approximieren können.

Chapter 3

Methodology

3.1 Environment Simulation

Eucledian simulation (discrete timesteps) Grid-based, every cell has environment state

- Gases amounts
 - O_2
 - CO_2
 - CH_4
 - H_2O
 - ...
- Temperature
- Light
- ...

There are global environment parameters such as:

- Gravity
- Length of day and night
- Axis tilt
- ...

3.1.1 Cellular Automata

Every cell's environment state is updated based on the state of the neighboring cells. Pressure equalization is simulated by diffusion of the gases.

3.2 Agent Representation

Every agent represents a single species/population. It has different attributes:

- ...

A cell holds a list of species and their amounts present in the cell.

3.2.1 Definition of a single species

Evolutionary lineage with similar enough traits to meaningfully differ from other species.

3.3 Approaches to Agent Control

How to update the population's attributes, specifically amount present and evolution over time? Good question... further research needed.

3.4 Self-Supervised *MARL*

Wenn biologische Organismen in eine völlig neue Umgebung geworfen werden, haben sie keinen zentralen "Score", den sie optimieren. Sie werden durch intrinsische Reize gesteuert. In dezentralen *MARL*-Systemen lässt sich das abbilden, indem man Agenten nicht für ein externes Ziel belohnt, sondern für die Informationsverarbeitung.

- Curiosity (Neugier) und Vorhersagefehler: Der Agent erhält eine Belohnung (Forward Prediction Loss), wenn er Zustände mit hoher Unsicherheit erkundet oder wenn er lernt, die Konsequenzen seiner Aktionen besser vorherzusagen. Er lernt also kontinuierlich, wie die Umwelt funktioniert, wodurch sich flexible Überlebensstrategien von ganz allein entwickeln.
- Sozialer Einfluss (Social Influence): Agenten werden dafür belohnt, dass ihre Aktionen das Verhalten oder den Zustand anderer Spezies vorhersehbar beeinflussen. Dies fördert das spontane Entstehen von realistischen inter-spezifischen Rollen (wie Prädator, Beute oder Symbiont), da die Spezies lernen, aufeinander zu reagieren, um ein soziales beziehungsweise ökologisches Gleichgewicht zu finden.

Chapter 4

Results

4.1 Rust Implementation

4.2 Performance

4.3 Accuracy

4.4 Comparison

Chapter 5

Conclusion

Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magnam aliquam quaerat voluptatem. Ut enim aequaleamus animo, cum corpore dolemus, fieri.

5.1 Limitations

5.2 Future Work

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Appendix A

External Resources

- GitHub Repository: <https://github.com/TimerErTim/bioinf.bachelor-thesis>
- Latorre et al. (2025) Data Repository: <https://github.com/rlatorre-uam/EcoEvoData>
- PredPreyGrass: <https://github.com/doesburg11/PredPreyGrass>

Appendix B

Source Codes

List of source codes:

B.1 Snippets

```
int main() {  
    return 0;  
}
```

References

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Abbreviations

Abbreviation	Description
AI	Artificial Intelligence
TaNa	Tangled Nature
MARL	Multi-Agent Reinforcement Learning
ABM	Agent-Based Modeling